

Couette Shear on the Infinite Periodic Channel: Exact Fourier Semigroup, Sharp Enhanced Dissipation, and Every Boundary

Route-A structural certificate C206

Abstract

For every real shear a , viscosity $\nu \geq 0$, and physical time, we derive the complete Fourier semigroup of scalar Couette advection–diffusion on $\mathbb{T} \times \mathbb{R}$. Completing one exact quadratic integral gives the exact and sharp k -sector norm $\exp\{-\nu(k^2t + a^2k^2t^3/12)\}$ and its cubic enhanced-dissipation scale, together with the precise norm-attainment boundary. We prove composition, inviscid unitary mixing, all no-shear/zero-mode/zero-time boundaries, the full L^2 periodic-state classification, and noncompactness excluding an ordinary Fredholm determinant. Finite exact cells audit conventions but do not prove the continuous theorem. The inviscid unitary is only a formal lift: all arithmetic and determinant gates fail, so Route A is rejected.

Keywords: Couette flow; shear mixing; enhanced dissipation; Fourier semigroup; noncompactness.

中文摘要

本文对无限周期通道上的Couette对流扩散方程给出全部实剪切、非负黏性与物理时间下的精确傅里叶半群。一次平方配方得到精确且尖锐的扇区范数、范数取等边界和三次增强耗散尺度，并统一处理无黏、无剪切、零模、零时刻、周期态及非迹类边界。有限证书只校验约定；由于不存在内生素数载体与目标行列式，路线A仍被拒绝。

1 Frozen equation and Fourier sign

Let $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ and consider

$$\partial_t f + ay\partial_x f = \nu(\partial_x^2 + \partial_y^2)f, \quad (x, y) \in \mathbb{T} \times \mathbb{R}, \quad a \in \mathbb{R}, \quad \nu \geq 0. \quad (1)$$

The clock is physical PDE time. We freeze

$$\widehat{f}_k(\eta) = \frac{1}{2\pi} \int_{\mathbb{T}} \int_{\mathbb{R}} f(x, y) e^{-i(kx + \eta y)} dy dx.$$

Since multiplication by y becomes $i\partial_\eta$, the transformed equation is

$$\partial_t \widehat{f}_k - ak\partial_\eta \widehat{f}_k = -\nu(k^2 + \eta^2)\widehat{f}_k. \quad (2)$$

Couette damping and enhanced-dissipation context is classical [1]; we claim a convention-complete derivation and certificate, not priority.

2 Exact semigroup and composition

Theorem 1. For every $k \in \mathbb{Z}$, $\eta \in \mathbb{R}$, and $t \geq 0$,

$$\widehat{S}_t \widehat{f}_k(\eta) = \exp \left[-\nu \left\{ k^2 t + t \left(\eta + \frac{akt}{2} \right)^2 + \frac{a^2 k^2 t^3}{12} \right\} \right] \widehat{f}_k(\eta + akt). \quad (3)$$

Moreover $S_t S_s = S_{t+s}$ on $L^2(\mathbb{T} \times \mathbb{R})$.

Proof. The characteristic ending at (t, η) is $\eta(r) = \eta + ak(t - r)$. Integrating (2) gives

$$D_t(\eta) = k^2 t + \int_0^t (\eta + akr)^2 dr = k^2 t + \eta^2 t + ak\eta t^2 + \frac{a^2 k^2 t^3}{3}.$$

Completing the square yields (3). Direct expansion gives $D_t(\eta) + D_s(\eta + akt) = D_{t+s}(\eta)$ and the shifts add, proving composition. $\square$

3 Sharp sector norm and scale

Proposition 1. On the k -th Fourier sector,

$$\|S_t\|_{k \rightarrow k} = \exp \left[-\nu \left(k^2 t + \frac{a^2 k^2 t^3}{12} \right) \right]. \quad (4)$$

If $\nu t > 0$, no nonzero L^2 vector attains this norm; frequency-localized packets approach it. If $\nu t = 0$, every nonzero vector attains the norm.

Proof. Write the sector operator as $\mathcal{M}_{m_t}U_t$, where $U_tg(\eta) = g(\eta + akt)$ is unitary and $m_t(\eta)$ is the positive exponential in (3). Its essential supremum is the value in (4): continuity and the unique quadratic minimizer $\eta_0 = -akt/2$ give the upper bound, while normalized functions supported in shrinking intervals about η_0 approach equality.

Let N denote the value in (4). If $\nu t > 0$ and a nonzero $g \in L^2(\mathbb{R}_\eta)$ attained the norm, equality of squared norms would give

$$\int_{\mathbb{R}} (N^2 - |m_t(\eta)|^2) |U_tg(\eta)|^2 d\eta = 0.$$

The first factor is strictly positive away from the singleton $\{\eta_0\}$, so U_tg would be supported on a null set and hence vanish, a contradiction. If $\nu t = 0$, $m_t \equiv 1$ and $\mathcal{M}_{m_t}U_t$ is unitary (the identity when $t = 0$), so every nonzero vector attains its norm. $\square$

For $a\nu k \neq 0$, the cubic factor reaches order one at $t = (12/(\nu a^2 k^2))^{1/3}$; the ordinary heat term $\nu k^2 t$ remains.

Revision focus. Round 0 freezes the Fourier convention and proves the exact semigroup, composition, sharp norm, and cubic scale.

Declarations

Data and code availability. Package-local exact synthetic evidence and deterministic code accompany the paper; no observational data are used.

Ethics. No human, animal, personal, or sensitive data are used.

Author contributions (CRediT). This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing. AI systems are not authors.

Funding. No external funding is reported.

Conflicts of interest. None known.

AI-use disclosure. An AI coding assistant supported derivation, drafting, and exact-code development; it was not an external reviewer or an independent peer-review process.

References

- [1] J. Bedrossian, N. Masmoudi, and V. Vicol, “Enhanced dissipation and inviscid damping in the inviscid limit of the Navier–Stokes equations near the two dimensional Couette flow,” *Arch. Rational Mech. Anal.* 219 (2016), 1087–1159. DOI: 10.1007/s00205-015-0917-3.